SCOPE Retrofit Measure Analysis Report

Comprehensive Energy and Financial Analysis

Executive Summary

This report compares energy consumption and operational costs between baseline and all applied renovation scenarios from CSV results. Embodied carbon is annualized using an analysis period of 30 years, while operational carbon is reported over 1 year. The construction cost data is from 2026 RSMeans Database. The embodied carbon is calculated using data from EC3 Environmental Product Declaration (EPD) Database. Since RSMeans Database is proprietary, users can adopt customized cost dataset when needed.

Building Information

- Weather File: USA_NY_Buffalo-Greater.Buffalo.Intl.AP.725280_TMY3.epw
- Building Name: SmallOffice
- Building Location: Buffalo, NY
- Total Floor Area: 511 m^2

Renovation Details by Scenario

Scenario	Renovation Type	Renovation Details
Baseline	baseline	Baseline (no envelope renovation)
Scenario 1	wall + roof + window + door	**Wall upgrade:** • Exterior wall insulation: Fiberglass Batts, target R-13.0 **Roof upgrade:** • Roof insulation: Blown Cellulose, target R-30.0 **Window upgrade:**

		• 2-pane replacement (skipped: SimpleGlazing windows) • Weatherstrip application (skipped: no OperableWindow) • Glazing film application (skipped: SimpleGlazing windows) • Caulking application: acrylic **Door upgrade:** • Door replacement: wooden door (1 of 3 door(s) skipped: door type incompatibility) • Bottom seal: automatic door bottom • Top/side seal: jamb weatherstrip

Annual Energy Analysis

Scenario	Metric	Baseline	Renovation	Delta	Savings %
Scenario 1	Total Site Energy (GJ)	440.31	423.91	**16.40**	**3.72%**

Key Performance Metrics

Operational Cost Saving (Scenario - Baseline) ($/yr) Comparison

Baseline $18,286

Scenario 1 $17,954

Baseline Scenario 1

Max saving: **$-332 (-1.82%)**

Operational Carbon Saving (Scenario - Baseline) (kg CO_2e/yr) Comparison

Baseline 40,219

Scenario 1 39,243

Baseline Scenario 1

Max saving: **-976 kg CO_2e (-2.43%)**

Min Retrofit Construction Cost

$11,612

Scenario 1

Min Retrofit Embodied Carbon

2,146 kg CO_2e

Embodied carbon intensity (ECI): 4.20 kgCO_2e/m^2

Scenario 1

Lowest Retrofit Cost Payback Period

35 years

Scenario 1

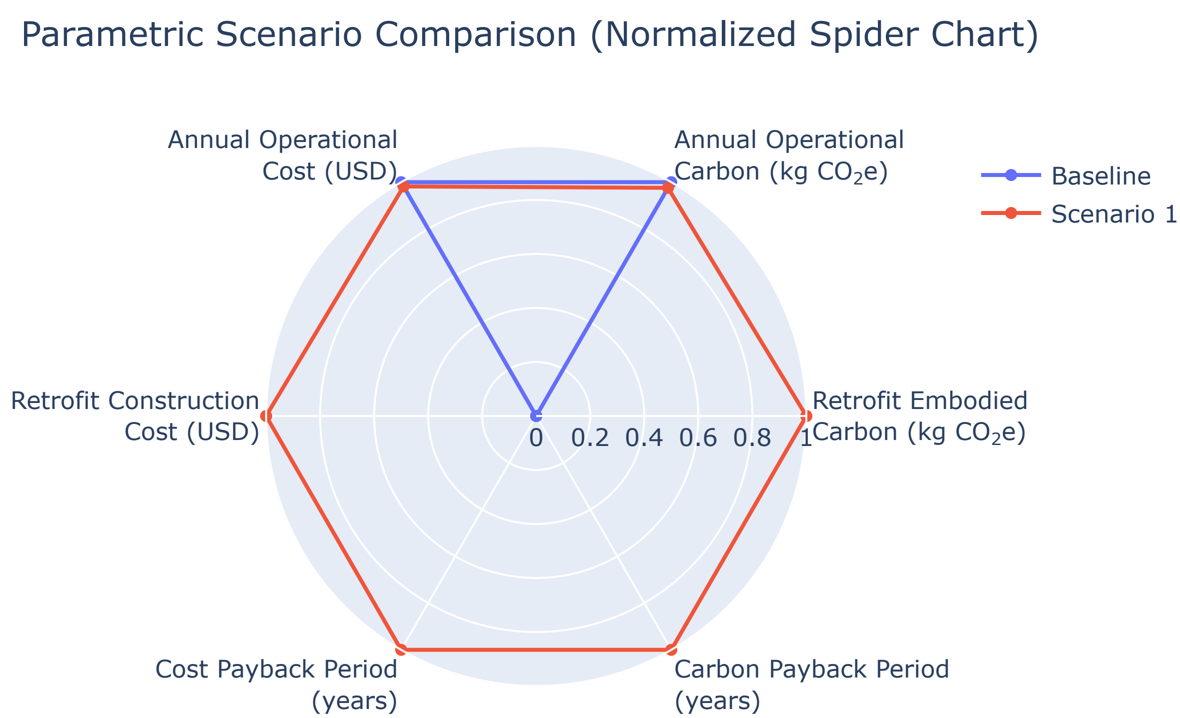
Lowest Retrofit Carbon Payback Period

2.20 years

Scenario 1

Comparative Visualizations between Renovation Scenarios

Spider Chart Visualization



Payback Period of the Retrofit

Cost Payback Period

Payback = retrofit construction cost / abs(Operational Cost Saving (Scenario - Baseline) ($/yr)).

Scenario 1 35 yrs

Carbon Payback Period

Payback = retrofit embodied carbon / abs(Operational Carbon Saving (Scenario - Baseline) (kg CO_2e/yr)).

Scenario 1 2.20 yrs

Material List

Material properties and consumption by renovation scenario. Missing values are shown as N/A.

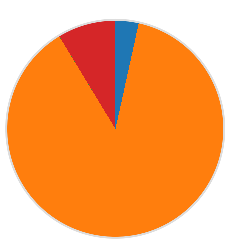
Scenario	Component	Material name	Volume (m^3)	Area (m^2)	Thickness (m)	Length (m)	Thermal Conductivity (W/m·K)	Density (kg/m^3)	Product Lifetime (years)
Scenario 1	Wall insulation	Fiberglass Batts	4.88	282	N/A	N/A	0.03	30	30
Scenario 1	Roof insulation	Blown Cellulose	125	599	0.21	N/A	0.04	50	30
Scenario 1	Window frame	N/A	N/A	6.51	N/A	N/A	N/A	N/A	15
Scenario 1	Window caulking	acrylic	0.0034	N/A	N/A	N/A	N/A	N/A	10
Scenario 1	Door	wooden door	N/A	3.90	N/A	N/A	0.15	638	30
Scenario 1	Door bottom sealing	automatic door bottom	N/A	N/A	N/A	3.66	N/A	N/A	15
Scenario 1	Door top/side sealing	jamb weatherstrip	N/A	N/A	N/A	16	N/A	N/A	15

Result Summary

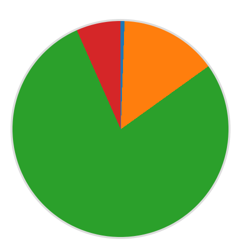
Per-scenario retrofit contribution breakdown. Each scenario includes two pie charts: cost and embodied carbon by retrofit measure.

Scenario 1

Cost breakdown by retrofit measure



Carbon breakdown by retrofit measure



- Wall: $400 (3.4%)
- Roof: $10,203 (87.9%)
- Window: $1.87 (0.0%)
- Door: $1,008 (8.7%)

Total: $11,612

- Wall: 12 (0.6%)
- Roof: 312 (14.5%)
- Window: 1,680 (78.3%)
- Door: 143 (6.6%)

Total: 2,146 kg CO_2e

Scenario	Retrofit Embodied Carbon (kg CO_2e)	Retrofit Construction Cost (USD)	Annual Operational Carbon (kg CO_2e)	Annual Operational Cost (USD)
Baseline	0.00	$0.00	40,219	$18,286
Scenario 1	2,146	$11,612	39,243	$17,954

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